Problem Statement & Modeling Decisions

Problem Statement:

Traditional room management systems often rely on cameras or manual monitoring to determine whether a space is occupied. These approaches may raise privacy concerns or require unnecessary human effort. Without understanding room occupancy patterns, it can also be difficult to manage lighting, temperature, and energy consumption efficiently.

Our project addresses this problem by using environmental sensor data to detect room occupancy without cameras and analyze the factors and patterns associated with room usage.

Target:

`Occupancy` determines whether the room is occupied (`1`) or not occupied (`0`).

Problem Type:

Binary Classification — the model predicts one of two classes: occupied or not occupied.

Main Evaluation Metrics:

1. **Macro F1-score:** Calculates the F1-score separately for each class and averages the results equally. This helps evaluate performance across both occupied and unoccupied rooms despite class imbalance.
2. **Confusion Matrix:** Shows the number of correct and incorrect predictions for each class, helping us identify cases where the model misclassifies occupied rooms as unoccupied or vice versa.

Trade-offs Against the Main Evaluation Metric:

We use Macro F1-score to evaluate both classes equally, especially because our dataset is imbalanced. However, optimizing Macro F1-score involves trade-offs:

- **False Positives vs. False Negatives:** A false positive occurs when the model predicts that a room is occupied when it is empty, potentially wasting energy. A false negative occurs when the model predicts that a room is empty when it is occupied, potentially causing discomfort or inefficient room management.
- **Precision vs. Recall:** Improving the model's ability to detect occupied rooms (Recall) may increase false positives, while reducing false positives (improving Precision) may cause the model to miss some occupied rooms.
- **Balanced Performance vs. Overall Accuracy:** Macro F1-score gives equal importance to occupied and unoccupied rooms, but this does not necessarily maximize overall accuracy or minimize every type of error.

We will compare Logistic Regression and Random Forest using Macro F1-score and the Confusion Matrix, while also considering Occupied Recall to understand how effectively each model detects occupied rooms.